EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**Calling a C# function that returns a new balance automatically changes the caller's original int variable.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

## GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.





DO NOW — RECALL

# Programming paradigms: applied comparison

Name: _____ Class: _____ Date: _____

**1. How does OOP differ from a procedural program?**

**2. What does a local variable mean?**

**3. If a field is private-set, how can callers change it safely?**

**4. Why does `Deposit(balance,5);` not update the caller's int balance if only a return value is produced?**

DO NOW — CHECK / OBJECTIVES

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

### 1. Model answer

OOP groups state and behaviour into objects; procedural code is organised around procedures/functions.

### 2. Model answer

A variable accessible only within its declaration's scope.

### 3. Model answer

Through public methods/property logic defined by the class.

### 4. Model answer

The return is discarded and a value-type parameter is a copy; the caller must assign the returned result.

### Vocabulary — know these terms

refactoring — Restructuring code while preserving expected behaviour.

encapsulation — Controlled access to internal state.

instance — Individual object created from a class.

return value — Output returned to the caller.

side effect — Observable change beyond a returned result.





YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 3

Compare one maintenance advantage and one risk of procedural and OOP implementations for a library system.

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 1

Procedural AddFive(x) returns x+5. If caller x=2 discards the result, what is x?

### Question 2

Two objects have balances A=10, B=30. A.Deposit(7) and B.Deposit(2) run. Find both balances and describe why they differ.





NEW LEARNING

# Read, annotate and explain

### Procedural state

The caller passes an integer balance and explicitly stores the returned value.

### Object state

Account object A stores Balance internally and Deposit changes that object's state.

### Separate instances

Account B stays at 20 when Account A changes to 15; the objects have independent state.

**Explain one step in your own words, using the correct technical terms.**

I DO — WORKED EXAMPLE

# Follow the worked method

**Two accounts start at 8 and 20; only first account deposits 4. Trace both.**

### Predict

Account a's Balance becomes 12.

### Model response

Account b remains at 20 because its state is separate.





WE DO — GUIDED PRACTICE

# Try → Discuss → Improve